

These three topics are the practical toolkit that sits next to ARIMA: smoothing is how you **see and forecast a level** without writing an AR polynomial; Granger is how you ask whether **another series actually helps**; ACF/PACF is how you **read memory** and choose p, d, q (and the seasonal period m).

The charts below are simulated on purpose so the pattern is clean. Real data is messier; the rules stay the same.

7. Smoothing methods

7.1 Why smoothing exists

A raw series is signal plus noise. Smoothing tries to estimate the **current level** of the signal so that:

- the plot becomes readable (description),
- short-term forecasts have something stable to stand on,
- sudden jumps are visible against a quiet background (monitoring).

Smoothing is not a substitute for ARIMA. It is a different philosophy.

- ARIMA says: “write an equation for how y_t depends on past y and past shocks.”
- Smoothing says: “keep a running estimate of the level (and maybe the trend, and maybe the season), and update it every period with a simple weighted average of what just arrived.”

That second approach is older, faster, and often surprisingly hard to beat for business forecasting — especially Holt–Winters on seasonal retail or energy series.

Two costs you always pay:

1. **Lag.** A smoother cannot turn until the new behaviour has been seen for a while. The wider the window (or the smaller the weight on the latest point), the later the turn.
 2. **Over-smooth vs under-smooth.** Too much smoothing hides real changes. Too little smoothing leaves you staring at noise.
-

7.2 Moving averages

All moving averages replace y_t by a local average of nearby points. They differ only in **which points get how much weight**.

Simple moving average — SMA

Last w observations, equal weight:

$$\text{SMA}_t(w) = \frac{y_t + y_{t-1} + \cdots + y_{t-w+1}}{w}$$

Characteristics:

- Easy to explain.
- Every point inside the window matters the same. A value from w days ago is treated like today's value.
- When a new point arrives, the oldest point is dropped. That can create odd kinks (the “drop-off” effect).
- The first $w - 1$ points have no SMA.
- Larger $w \rightarrow$ smoother, more lag. On monthly data, $w = 12$ is a crude seasonal average; it will also flatten a genuine trend.

Use SMA when you want a transparent rolling level, not when you want a serious forecast.

Weighted moving average — WMA

Same window, but recent points get larger weights. A common linear scheme:

$$\text{WMA}_t(w) = \frac{w y_t + (w - 1) y_{t-1} + \cdots + 1 \cdot y_{t-w+1}}{w + (w - 1) + \cdots + 1}$$

Characteristics:

- Still a finite window.
- Reacts faster than SMA of the same width, because yesterday counts more than the oldest point.
- You must choose both the window and the weight pattern. That is two knobs instead of one.

Exponential moving average — EMA

Now the window is **infinite in theory**, but weights die exponentially as you go backwards.

Let $\alpha \in (0, 1]$ be the weight on the newest observation. Then

$$\text{EMA}_t = \alpha y_t + (1 - \alpha) \text{EMA}_{t-1}$$

Unrolled:

$$\text{EMA}_t = \alpha y_t + \alpha(1 - \alpha)y_{t-1} + \alpha(1 - \alpha)^2 y_{t-2} + \dots$$

The weights $\alpha, \alpha(1 - \alpha), \alpha(1 - \alpha)^2, \dots$ sum to 1. Nothing is ever fully dropped; old data just fades.

People also parameterise EMA by a **span** s , related to α by

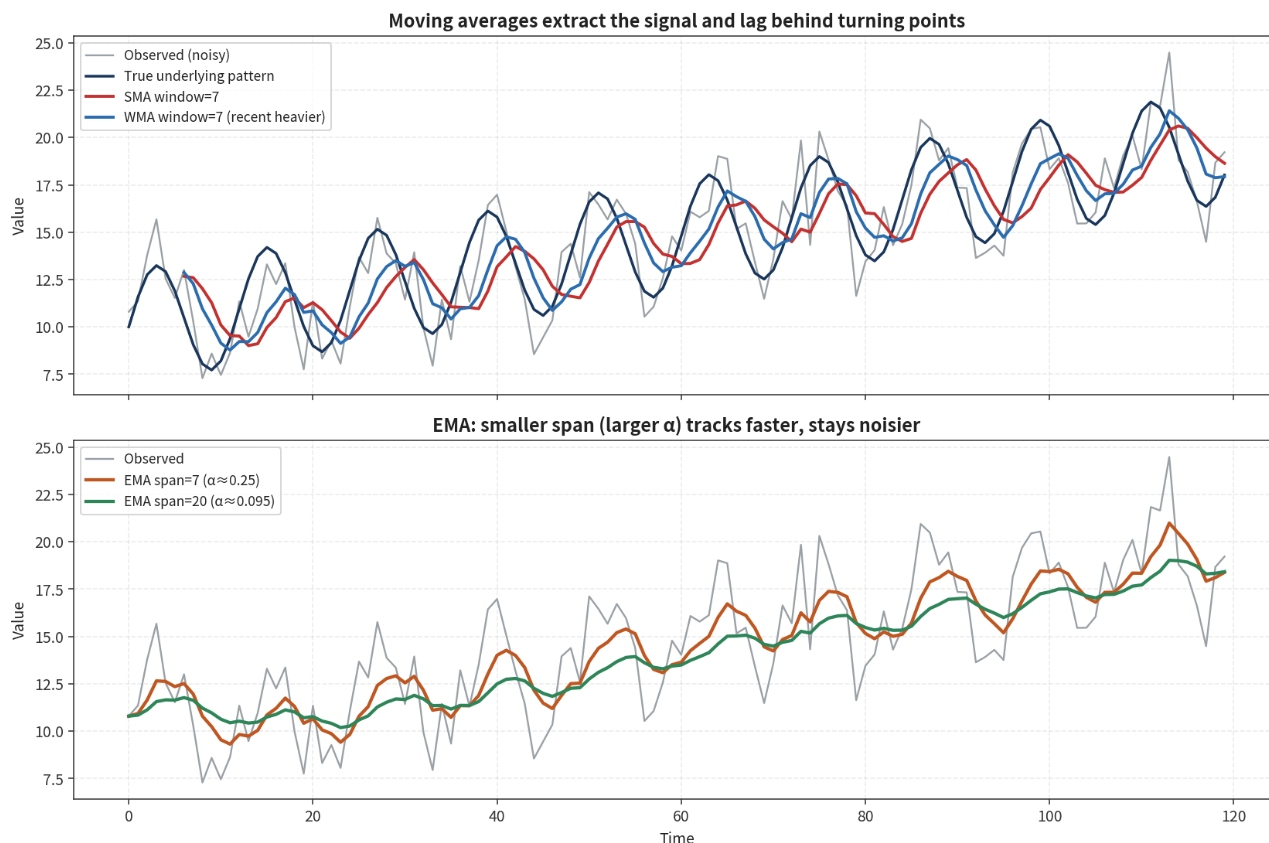
$$\alpha = \frac{2}{s + 1}$$

So span 7 gives $\alpha \approx 0.25$; span 20 gives $\alpha \approx 0.095$.

Characteristics:

- One parameter, α .
- Reacts immediately (the newest point always gets weight α).
- No hard cutoff of old history.
- This is the same recursion that **single exponential smoothing** uses. That is the origin of the “EMA vs ES” confusion, which we will settle below.

How these look on a noisy series with a slow rise and a yearly wiggle:



Read that figure as:

- SMA and WMA of width 7 both recover the shape and both lag a little at turning points.
- WMA hugs turns slightly faster than SMA.
- EMA with a large α (small span) follows the series tightly and stays noisy.
- EMA with a small α (large span) is calm and late.

There is no universally correct width. You choose it from the frequency of the series and from how fast the world actually changes.

7.3 Exponential smoothing (the forecasting family)

Moving averages are usually used as **filters**: they produce a smoother version of the past.

Exponential smoothing, in the Holt–Winters sense, is a **forecasting system**. It keeps a small set of **state variables** and updates them each period.

The one parameter you must understand: α

In single exponential smoothing the level L_t is updated by

$$L_t = \alpha y_t + (1 - \alpha)L_{t-1}$$

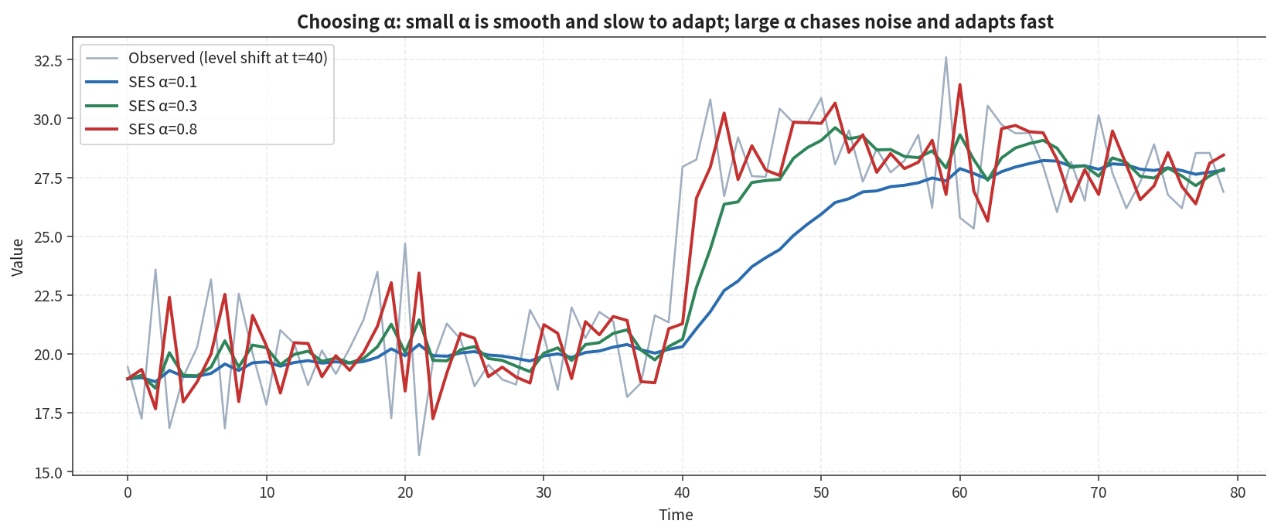
and the forecast for every future horizon is just that level:

$$\hat{y}_{t+h} = L_t \quad h = 1, 2, 3, \dots$$

What α does:

- α near 0: almost ignore the new observation. The level moves like a tanker. Good when the series is very noisy and the true level changes slowly.
- α near 1: treat the new observation as the new truth. The level jumps around. Good when the series really did change and noise is small.
- Typical fitted values on business data sit somewhere between 0.1 and 0.4, but that is a tendency, not a law.

Same series, a sudden permanent jump at $t = 40$, three different α :



Small α takes a long time to accept the new level. Large α accepts it quickly and also rides the noise before and after the jump. In software, α is usually **estimated by minimising one-step forecast error** (SSE or likelihood) rather than picked by hand — but you should still look at the fitted value. An α of 0.99 means the method has decided the series is almost a random walk. An α of 0.02 means it has decided almost nothing new is information.

Single exponential smoothing (SES)

State: level only.

$$L_t = \alpha y_t + (1 - \alpha)L_{t-1}$$
$$\hat{y}_{t+h} = L_t$$

Use it when there is no trend and no seasonality — a stationary-looking series that jitters around a slowly changing mean.

Forecast shape: a **horizontal line**. SES cannot predict growth. If you use it on a trending series, the forecasts sit systematically below (or above) the future path.

Double exponential smoothing / Holt's linear trend

State: level + slope.

Two smoothing constants, α for the level and β for the trend:

$$L_t = \alpha y_t + (1 - \alpha)(L_{t-1} + T_{t-1})$$
$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$$
$$\hat{y}_{t+h} = L_t + hT_t$$

Read the first equation as: the new level is a blend of the new observation and **where we expected to be** (old level plus old slope).

Read the second as: the new slope is a blend of the just-observed change in level and the old slope.

Forecast shape: a **straight line** continuing the latest slope. Good for series with trend and no season. Bad if the trend will not continue (you will fly off into space), and bad if there is a yearly wave (the wave is ignored).

There is a damped-trend variant, Holt with damping $\phi < 1$:

$$\hat{y}_{t+h} = L_t + (\phi + \phi^2 + \dots + \phi^h)T_t$$

The slope is allowed to fade. This is often more honest for long horizons.

Triple exponential smoothing / Holt–Winters

State: level + slope + seasonal factors.

Two flavours, matching the additive / multiplicative choice from decomposition.

Additive Holt–Winters (seasonal swing has constant size):

$$\begin{aligned}L_t &= \alpha(y_t - S_{t-m}) + (1 - \alpha)(L_{t-1} + T_{t-1}) \\T_t &= \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \\S_t &= \gamma(y_t - L_t) + (1 - \gamma)S_{t-m} \\\hat{y}_{t+h} &= L_t + hT_t + S_{t-m+1+((h-1) \bmod m)}\end{aligned}$$

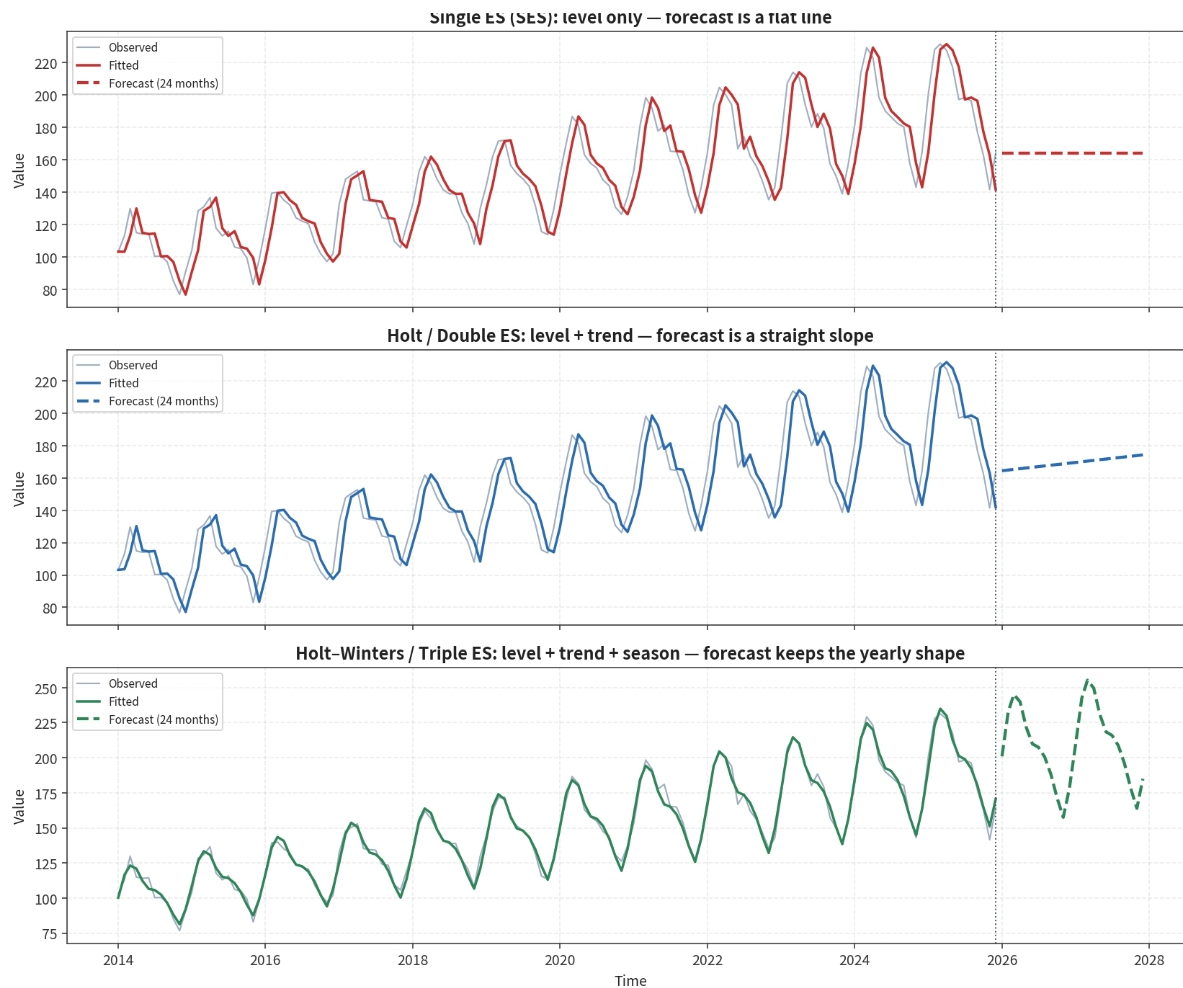
Multiplicative Holt–Winters (seasonal swing scales with the level — typical of growing sales):

the seasonal piece multiplies rather than adds, and the update of L_t divides by last year's seasonal factor.

Here m is the seasonal period (12 for months, 4 for quarters, 7 for a weekly pattern in daily data). γ is the seasonal smoothing constant: how fast the January factor is allowed to change from last January.

Forecast shape: the trend continues, and the seasonal wave is replayed.

The three methods on one growing seasonal series:



That figure is the whole family in one glance.

- SES: fitted line is a delayed level; forecast is flat. It cannot keep up with growth or with July peaks.
- Holt: fitted line follows the rise; forecast is a slope with the seasonal wave stripped out.
- Holt–Winters: fitted line tracks both rise and season; forecast continues both.

If you remember only one practical rule: **match the method to the components you can see**.
No trend, no season → SES. Trend, no season → Holt. Trend and season → Holt–Winters. If the seasonal waves get taller as the series grows, use multiplicative season.

7.4 EMA versus exponential smoothing — they are not two different worlds

This bullet on the slide exists because the names collide.

EMA (G) is a special case of SES (SES) with $\alpha = 2/(1 + 2\lambda)$ and $\lambda = 1/(2\lambda + 1)$.

EMA (in finance and in pandas `ewm`) is a **filter**. You compute a smoothed version of the observed series. You do not, by itself, produce a 12-month forecast with a trend and a season. One parameter, α or span.

SES is EMA used as a **forecasting model**. The EMA value *is* the level, and the forecast is that level, held constant. Same recursion.

Holt and Holt–Winters generalise that recursion by adding extra state variables (slope, seasonal factors) and extra smoothing constants (β, γ).

So:

- Every SES level is an EMA of a carefully chosen residual (the deviation from the previous forecast).
 - Not every EMA is a full exponential-smoothing forecasting system.
 - When someone says “we used exponential smoothing,” ask: SES, Holt, or Holt–Winters? Additive or multiplicative? What m ?
-

8. Granger causality

Granger causality is one of the most misquoted ideas in applied work, so we will be precise.

What the test is actually asking

Take two series, X and Y .

We say X **Granger-causes** Y if past values of X help you forecast Y **beyond what past values of Y already do**.

It is a statement about **predictive content in time**, not about true physical or moral cause.

The usual implementation is a pair of regressions (this is the VAR idea from the previous section, written as two equations).

Restricted model — Y explains itself:

$$y_t = a_0 + a_1 y_{t-1} + \cdots + a_p y_{t-p} + u_t$$

Unrestricted model — Y is allowed to use lags of X as well:

$$y_t = a_0 + a_1 y_{t-1} + \cdots + a_p y_{t-p} + b_1 x_{t-1} + \cdots + b_p x_{t-p} + e_t$$

Null hypothesis:

$$H_0 : b_1 = b_2 = \cdots = b_p = 0$$

If you reject H_0 , past X improved the forecast of Y . People then say “ X Granger-causes Y .”

You should always run the test **both ways**. It is entirely possible that X helps Y and Y helps X (feedback), or that neither helps the other.

What it is not

- It is not “ X caused Y in the real world.” Rain Granger-causes umbrella sales. Umbrella sales also often Granger-cause rain readings if shops stock umbrellas just before the monsoon and your weather series is noisy. The test cannot see the mechanism.
- It is not valid, as usually taught, on raw non-stationary series. If both series are random walks, they can look related just because both wander. Standard practice: difference them to stationarity, *or* work in a cointegration / VECM setup if they share a long-run level.
- It depends on the lag length p . Too short, you miss delayed effects. Too long, you waste degrees of freedom. People use AIC/BIC or a rule tied to frequency (for quarterly data, $p = 4$ or 8 is a common starting point).
- It assumes you have not left out a third series Z that drives both. If oil prices drive both inflation and the policy rate, a two-variable test can lie.

How to read a result in words

Suppose monthly advertising spend A_t and monthly sales S_t , both made stationary, $p = 3$.

- Reject “ A does not Granger-cause S ”, fail to reject the reverse: advertising helps predict sales, sales do not help predict advertising. That is the pattern you *hope* for in a campaign.
- Reject both directions: the two series forecast each other. Maybe good campaigns follow good sales (budget rules), and also lift later sales.

- Fail to reject both: knowing one does not improve the short-term forecast of the other, at least at these lags.

Granger belongs with VAR, not with univariate ARIMA. You need at least two series.

9. ACF and PACF

This is the reading skill the whole Box–Jenkins workflow rests on. We already used the rules when we discussed AR and MA. Here we make the functions themselves, and the identification table, fully explicit.

9.1 ACF — autocorrelation function

The autocorrelation at lag k is the ordinary correlation between the series and itself shifted by k steps:

$$\rho_k = \frac{\text{Cov}(y_t, y_{t-k})}{\text{Var}(y_t)}$$

The **ACF plot** is $\hat{\rho}_1, \hat{\rho}_2, \dots, \hat{\rho}_K$ as bars.

On a sample of length T , bars are usually compared to the white-noise band

$$\pm \frac{1.96}{\sqrt{T}}$$

A bar inside that band is compatible with “this lag is noise.” A bar outside is a lag worth taking seriously — with the usual warning that if you look at 40 lags you should expect one or two false spikes.

What ACF measures: **total** linear memory at lag k , including memory that merely travels through the in-between lags. If y_t depends on y_{t-1} and y_{t-1} depends on y_{t-2} , then ρ_2 will not be zero even if there is no *direct* two-step effect.

9.2 PACF — partial autocorrelation function

The partial autocorrelation at lag k is the correlation between y_t and y_{t-k} **after the linear**

effects of $y_{t-1}, \dots, y_{t-k+1}$ have been removed.

One construction: regress y_t on lags 1 through k , and read the coefficient on lag k . That coefficient is $\hat{\phi}_{kk}$, the PACF at lag k .

What PACF measures: **direct** extra memory at lag k , over and above what the closer lags already explain.

That is why PACF is the natural tool for reading an AR order. An $AR(p)$ says: once you have included the last p values, further lags add nothing. So ϕ_{kk} should be visible up to $k = p$ and then fall inside the band.

9.3 The key difference

	ACF	PACF
Question it answers	How related are y_t and y_{t-k} , ignoring the middle?	How related are they after the middle has been partialled out?
Reads which model	MA order q (cutoff in ACF)	AR order p (cutoff in PACF)
On an $AR(p)$	Tails off (geometric / damped sine)	Cuts off after p
On an $MA(q)$	Cuts off after q	Tails off
On white noise	Dead after lag 0	Dead after lag 0
On a random walk / unit root	Very slow, almost linear decay	Large spike at lag 1, then messy

A memory aid that actually works:

PACF $\rightarrow$ P for $AR(p)$.

ACF $\rightarrow$ leftover name, so it carries the $MA(q)$.

“Cutoff” means: the function is clearly outside the band up to that lag, then falls inside and stays there.

“Tail off” means: it shrinks gradually across many lags, often looking like a geometric decay or a dying wave. It does not drop off a cliff.

9.4 Order of models: p, d, q, m

The slide writes (p, d, q, m) . In standard notation:

- p — non-seasonal AR order (how many past *values*)
- d — non-seasonal differences needed for stationarity
- q — non-seasonal MA order (how many past *shocks*)
- m or s — seasonal period (12, 7, 24, 4, ...)
- and, when you write SARIMA, also P, D, Q at that seasonal lag

You do **not** read p and q on a series that still has a wandering mean. The slow ACF of a random walk will tempt you to fit AR(12). Difference first, then read the plots again. That is d .

How d is chosen, in order:

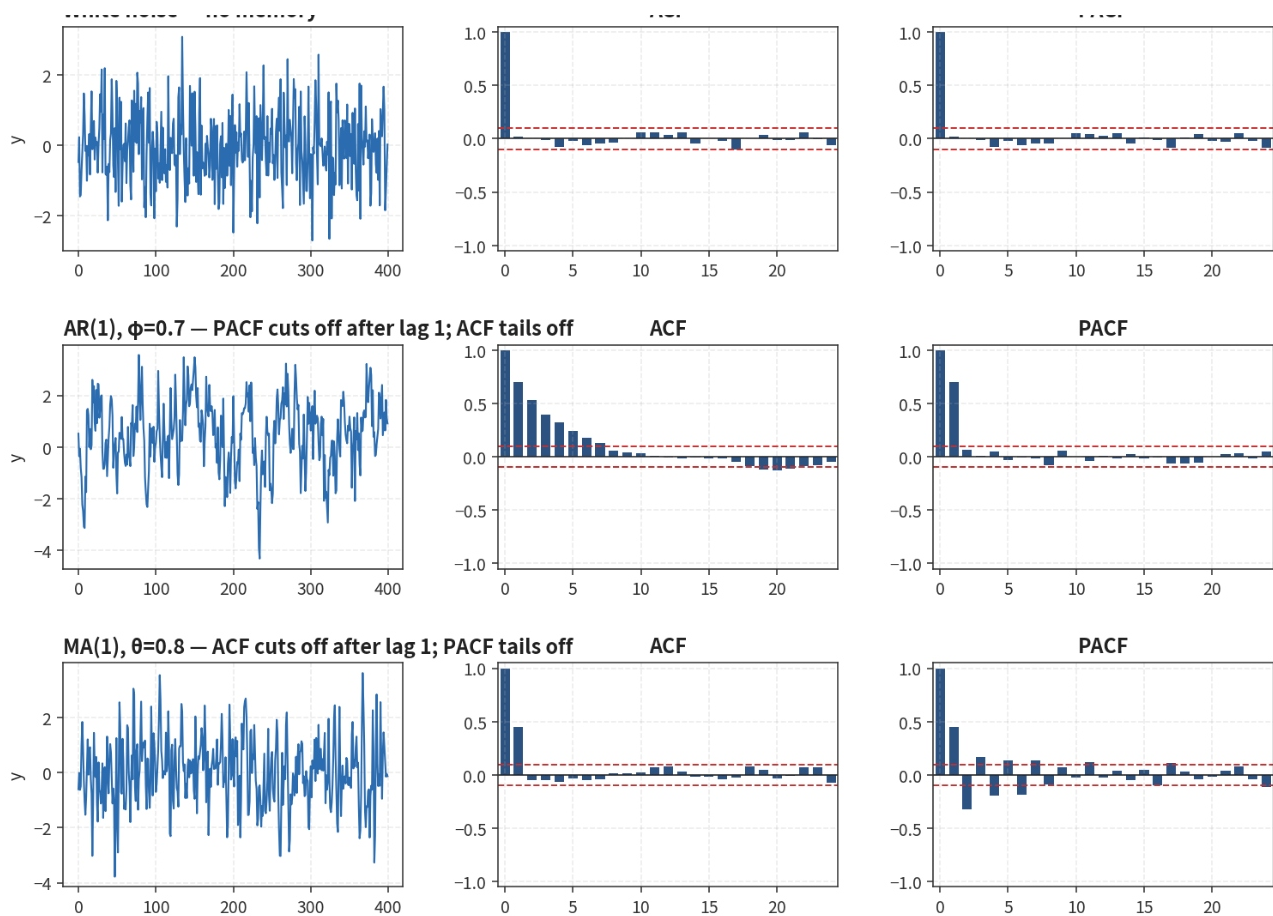
1. Look at the raw plot. Does the level wander?
2. Look at the raw ACF. Does it die very slowly?
3. ADF / KPSS.
4. Difference once, repeat. Stop when the differenced series looks stationary and the ACF no longer crawls down from 1.
5. Rarely go past $d = 2$. If you think you need $d = 3$, the series probably wants a log, a break dummy, or a different model.

How m is chosen: from the calendar and from the ACF. Monthly data starts with $m = 12$. Confirm by seeing spikes at 12, 24, 36. Daily data with a weekly rhythm has $m = 7$. If those seasonal spikes decay very slowly, you also need a seasonal difference $D = 1$, i.e. $y_t - y_{t-m}$.

9.5 Identifying models from ACF and PACF

Below, each row is one process: the series itself, its ACF, its PACF. Red dashed lines are the white-noise bands.

White noise, AR(1), MA(1):

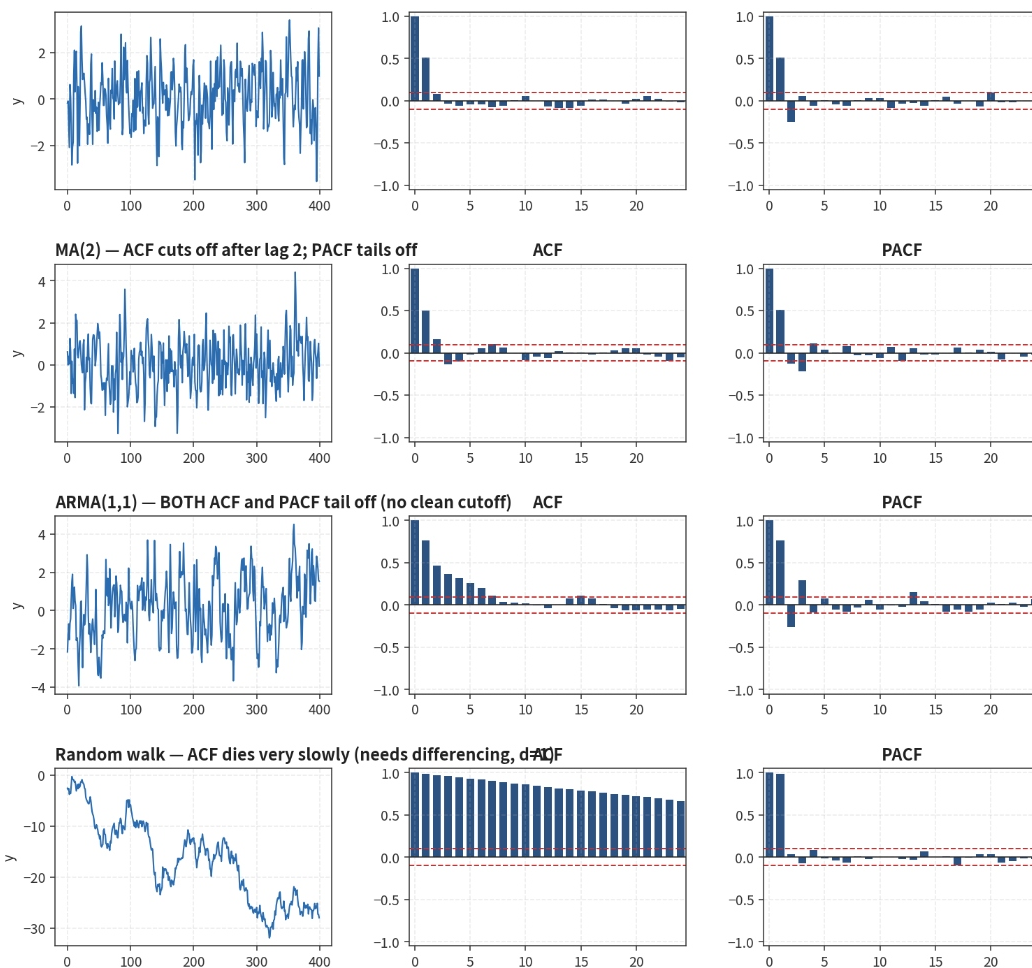


AR(2), MA(2), ARMA(1,1), random walk:

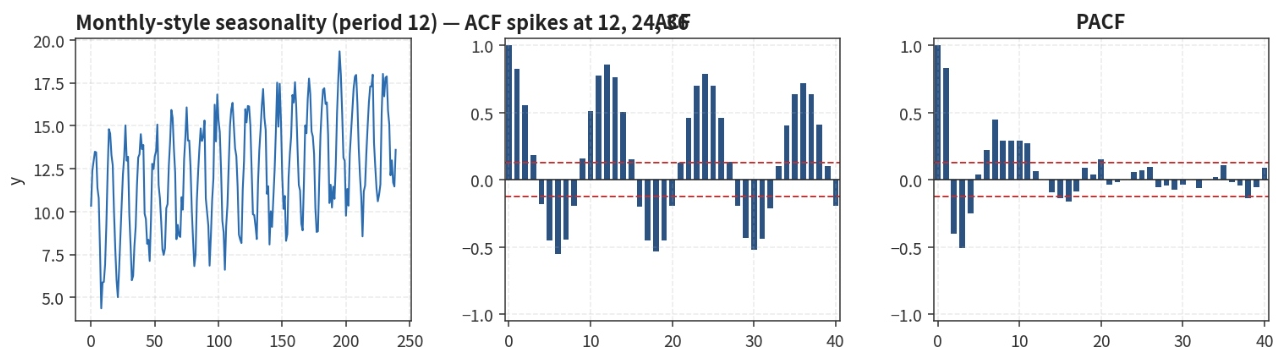
AR(2) — PACF cuts off after lag 2; ACF tails off

ACF

PACF



Seasonal period 12 (the extra spikes that tell you m):



Worked reading of those figures:

1. **White noise.** Both plots dead after lag 0. There is nothing to model. Forecast the mean.
2. **AR(1).** ACF: a run of decaying bars. PACF: one large bar at lag 1, then silence. Read $p = 1$, $q = 0$, and only after you already know $d = 0$.
3. **MA(1).** ACF: one large bar at lag 1, then silence. PACF: a decaying tail. Read $q = 1$, $p = 0$

4. **AR(2)**. PACF has two standing bars, then silence. ACF tails off (sometimes it oscillates if the AR polynomial has complex roots). Read $p = 2$.
5. **MA(2)**. ACF has two standing bars, then silence. PACF tails off. Read $q = 2$.
6. **ARMA(1,1)**. Both plots tail off. Do not force a cutoff that is not there. Try small mixed orders, compare AIC/BIC, check residuals.
7. **Random walk**. ACF starts near 1 and slides down for a long time. This is **not** an AR(20). This is “difference once.” After differencing you should get something close to white noise, which is ARIMA(0,1,0).
8. **Seasonal monthly series**. Extra ACF spikes at 12, 24, 36. That is $m = 12$. If those spikes themselves die slowly, take a seasonal difference. Then read P and Q off the seasonal lags the same way you read p and q off the short lags.

A compact identification table you can keep:

Pattern after the series is stationary	Model to try
ACF and PACF both dead	White noise
PACF cuts off at lag p ; ACF tails off	AR(p)
ACF cuts off at lag q ; PACF tails off	MA(q)
Both tail off	ARMA(p, q), search small values
ACF slow linear decay <i>before</i> differencing	Raise d (usually to 1) and redraw
Spikes at multiples of m	Seasonal terms; maybe $D = 1$
Residual ACF still has spikes after fitting	The order is wrong, or a season / break remains

Always close the loop: fit, then plot **residual** ACF/PACF. The residuals should look like the white-noise row. If lag 12 is still standing in the residual ACF, you did not finish the seasonal part.

How sections 7, 8 and 9 sit with what you already have

- Smoothing (Section 7) is a **forecasting family that does not require you to choose p and q** . Holt–Winters is the default competitor to SARIMA on seasonal business data.
- Granger (Section 8) is how you decide whether a second series deserves to enter a VAR / ARIMAX, rather than staying univariate.

- ACF/PACF (Section 9) is how you choose the univariate orders, and how you audit residuals of *both* ARIMA and smoothing models.

A reasonable working order on a new series:

1. Plot. Name trend, season, breaks.
 2. Decide smoothing versus ARIMA (or fit both and compare out-of-sample).
 3. If ARIMA: stationarity $\rightarrow d$ (and D) $\rightarrow$ ACF/PACF $\rightarrow p, q, P, Q \rightarrow$ residual ACF.
 4. If you have extra series and a “does this help?” question: Granger on stationary versions, then VAR / ARIMAX.
-

Resources that match this exact layer

- Hyndman & Athanasopoulos, *Forecasting: Principles and Practice* — the chapters on exponential smoothing and on ARIMA identification. Free online; the best written source for SES / Holt / Holt–Winters.
 - Box–Jenkins identification chapter in any standard time-series text (Brockwell & Davis, or Shumway & Stoffer) for ACF/PACF theory.
 - `statsmodels` in Python: `SimpleExpSmoothing`, `ExponentialSmoothing`, `plot_acf`, `plot_pacf`, `grangercausalitytests`.
 - For Granger caveats: any decent econometrics notes on “Granger causality is not causality.”
-

If you want to continue in this depth, the highest-value next pieces are:

1. **Section 4 stationarity** (ADF vs KPSS, what differencing actually does to ACF) — this is what justifies d .
2. **Section 2–3 decomposition vs Holt–Winters vs STL** — same components, three tools.
3. A **full worked SARIMA + Holt–Winters bake-off** on one series, with residual ACF shown side by side.

Save which of those you want next.